Space Debris and Satellite Detection

YOLO11n-Based Object Detection Using Deep Learning

Presented by:

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Introduction

- As machine learning (ML) continues to expand across various applications, computer vision (CV) has emerged as a prominent research area over the past several decades.
- Decades of human activity in space, including exploration, experimentation, and technological development, have resulted in the accumulation of unwanted objects in Earth's orbit.
- Orbital debris presents a substantial threat to the continued use and exploration of space.Such impacts can be catastrophic and may generate additional fragments, further increasing the quantity of debris in orbit.
- Reliable object detection can identify operational limitations or unsuitable targets at an early stage, thereby improving both mission efficiency and safety.
- The first is image degradation caused by interference from celestial objects and sensor noise, which can introduce substantial disturbances and reduce detection accuracy.
- An important step in this process is the detection of space debris, which subsequently enables its tracking and cataloging.
- Among these, visible-light sensors are particularly promising because of their high level of autonomy and accurate observational capabilities.
- Continuous monitoring of space debris and accurate analysis of its trajectory are essential for reliable detection, timely collision warnings, and the protection of operational spacecraft.
- Because the debris occupies only a very small region of the image, these disturbances can substantially reduce the signal-to-noise ratio, making reliable detection difficult.

Related Work

- A widely used CV application is object detection (OD), which combines image processing techniques with deep learning (DL) to identify and locate objects within visual data.
- This escalating chain of collisions is commonly associated with the Kessler Syndrome (Kessler & Cour-Palais, 1978), in which collisions produce additional debris that subsequently causes further collisions.
- Examples include light-curve data (Allworth et al., 2021), RGB images (AlDahoul et al., 2022), and radar observations (Mehrholz et al., 2002), with image-based data being among the most frequently used sources.
- Existing solutions generally concentrate on improvements to network architecture and data augmentation strategies
- From an architectural perspective, Feature Pyramid Networks (FPNs) and their variants are widely adopted because they facilitate the fusion of features at different spatial scales.
- Because one-stage detectors generally require only one forward pass through the network, they offer lower computational latency and faster inference than many two-stage approaches.
- YOLO has become one of the representative one-stage object detection frameworks because of its combination of detection accuracy and real-time inference capability.

Related Work

- Joseph Redmon continued to develop YOLOv3 based on YOLOv2 in 2018.
- Researchers such as Alexey Bochkovskiy proposed the YOLOv4 algorithm model based on YOLOv3 in 2020.
- The YOLOv5 algorithm model, which has undergone improvements and iterations, was subsequently released by Ultralytics.
- Alexey Bochkovskiy proposed the YOLOv7 algorithm in 2022, further improving the network's speed and accuracy by utilizing techniques like Efficient Aggregation Networks, dynamic label assignment, and re-parameterized convolution.

Methodology

- Dataset Collection
- Data Preprocessing & Annotation
- Train–Validation–Test Split
- YOLO Model Training
- Model Validation
- Testing & Detection
- Performance Evaluation

The proposed methodology uses a YOLO-based object detection approach to detect and localize satellites/space debris using bounding boxes

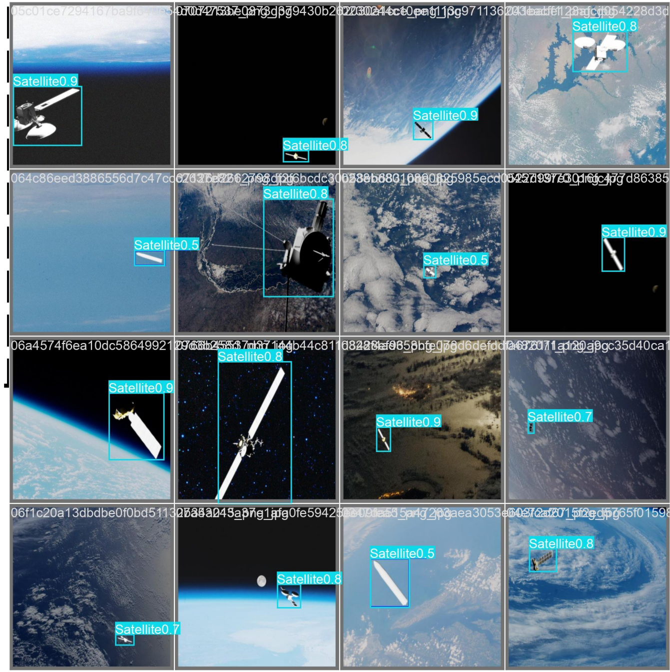
Dataset Preparation and Model Training

Dataset Preparation

- Collect satellite/space-debris images
- Annotate objects using bounding boxes
- Convert annotations to YOLO format
- Divide dataset into training, validation and testing sets

Model Training

- YOLO-based object detection
- Trained for **50 epochs**
- Learns object features and bounding-box locations
- Training monitored using box, classification and DFL losses



DATASET NAME : Space Debris Computer Vision Dataset by Capstone Design

DATASET LINK: https://universe.roboflow.com/capstone-design-ecfaa/space-debris-pidgc

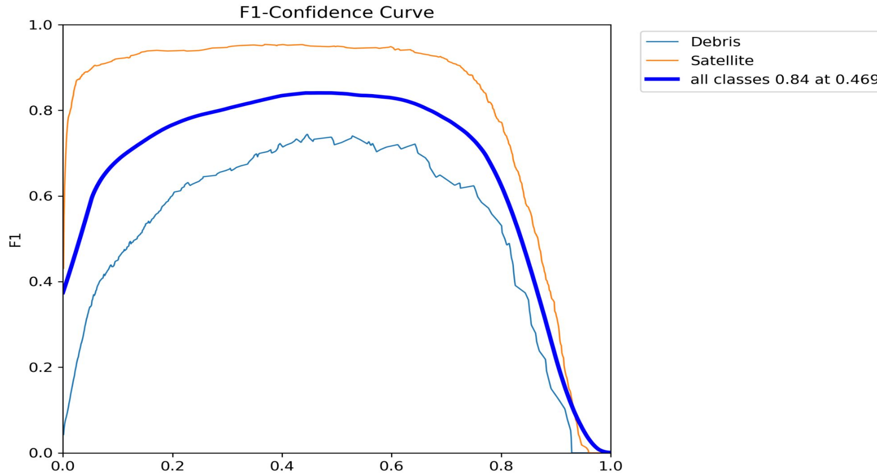
Model Validation and Performance Evaluation

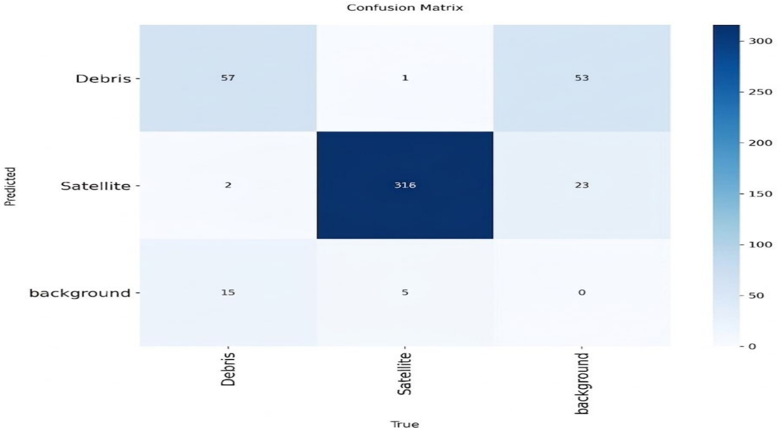
Validation
→ Evaluate model on validation images
→ Monitor detection performance

Testing & detection
→ Test and evaluate on unseen images
→ Generate bounding boxes + class labels + confidence score

Evaluation Metrics

- Precision
- Recall
- mAP@50
- mAP@50–95
- F1-score

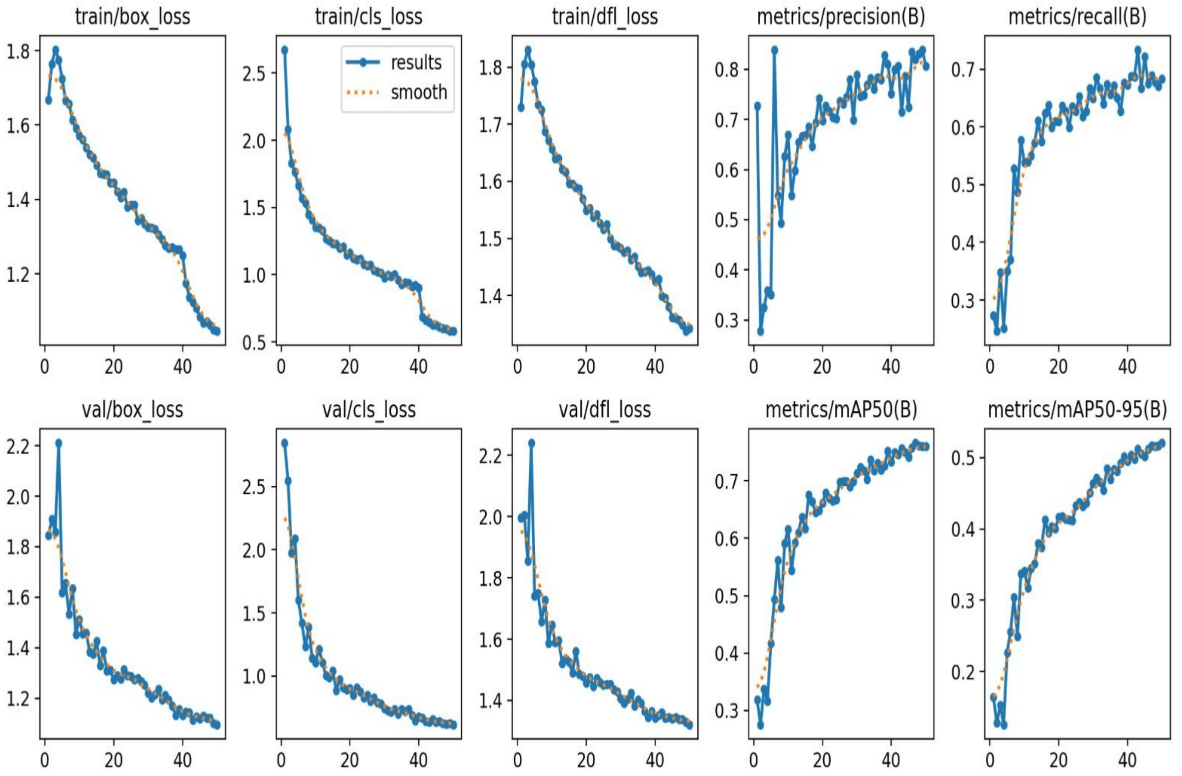




Results- Overall Model Performance

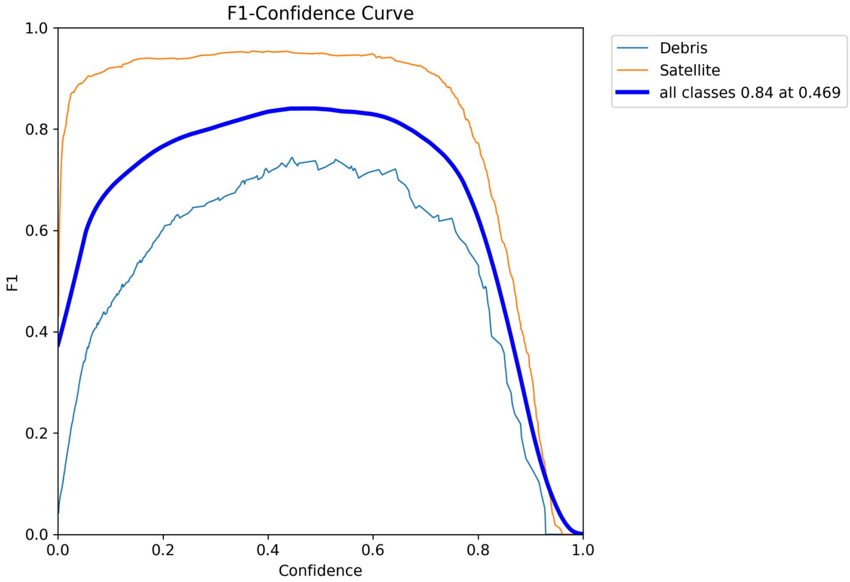
Yolo11n | 50 Epochs | 640 × 640 Input

Precision	Recall	mAP@50	mAP@50–95
80.8%	68.3%	76.0%	52.0%



Class-wise Detection Performance

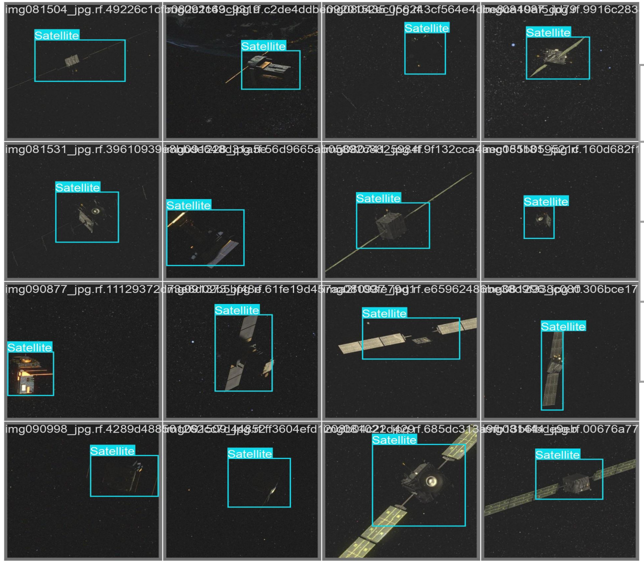
class	precision	Recall	mAP@50	mAP@50-95
Debris	**65.9%**	**43.2%**	**54.4%**	**32.3%**
Satelite	**95.7%**	**93.4%**	**97.6%**	**71.8%**



Best overall F1-score: 0.84 at confidence 0.469

Satellite detection performed significantly better than debris detection.

Sample Detection Results

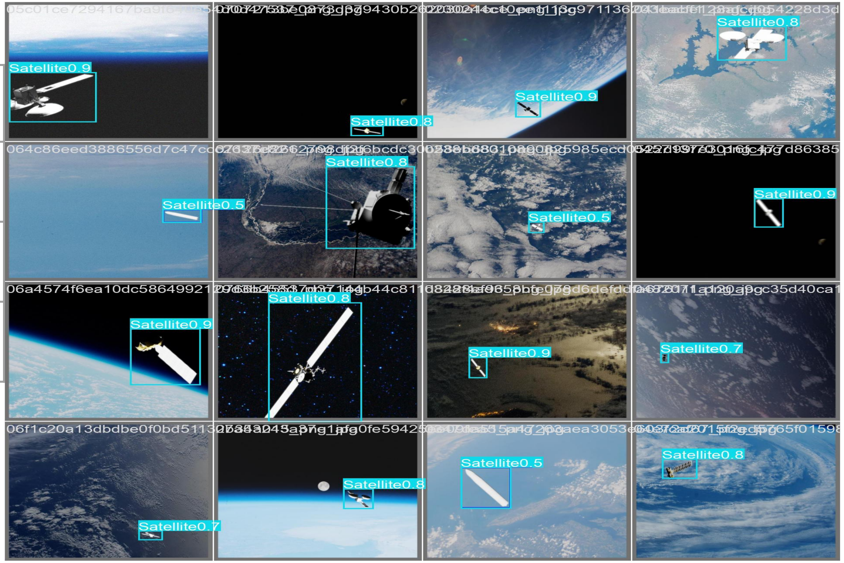


- Detection Output

1.Class Label

2.Bounding box

3.Confidence score



The trained model successfully detected and localized satellites and space debris using bounding boxes and confidence scores

Conclusions

YOLO11n successfully detected and localized satellites and space debris using bounding boxes

Key Limitation

Debris detection remains the main area requiring improvement.

Key Findings

- Precision: **80.8%**
- Recall: **68.3%**
- mAP@50: **76.0%**
- Best F1-score: **0.84**
- Satellite detection: **97.6% mAP@50**
- Debris detection: **54.4% mAP@50**

Future Scope

.Larger dataset
.Better debris
.representation Data
.augmentation Model
optimization
.Real-world testing

References

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